SOFTWARE ENGINEERING

LAB ASSIGNMENT 05

Sequence Diagram — Money Transfer

Digital Wallet Management System (DWMS)

SUBMITTED BY — TEAM 5

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1. Introduction to Sequence Diagrams

A UML Sequence Diagram is an interaction diagram that shows how actors and system objects exchange messages over time. Participants are placed across the top, vertical lifelines indicate their existence during the interaction, and horizontal arrows represent requests and responses. Sequence diagrams help verify that a business scenario follows the intended order and that data changes occur only after required checks.

1.1 Objectives

- Model the user-to-user money-transfer scenario described in the SRS.
- Show recipient/balance validation before confirmation and ledger updates.
- Represent atomic wallet updates and notifications to both parties.

2. Scenario and Assumptions

An authenticated sender specifies a recipient and amount. The DWMS validates that the recipient exists and the sender has sufficient funds, presents a summary, and waits for explicit confirmation. On confirmation, it updates the sender and recipient wallets atomically, writes the transaction/audit records, and issues a receipt and notifications.

2.1 Assumptions

- The sender is already authenticated, consistent with the precondition for wallet operations.
- The Wallet/Ledger DB represents the persistent data layer; internal schema and database vendor are not fixed by the SRS.
- The diagram models a successful transfer path; validation or commit failures must return an error and must not leave partial balance updates.

SRS Reference	Design interpretation
REQ-XFER-1	Capture recipient and transfer amount.
REQ-XFER-2	Validate recipient existence and sufficient sender balance.
REQ-XFER-3	Require explicit confirmation before committing.
REQ-XFER-4	Update sender and recipient balances atomically.
REQ-XFER-5	Record transfer in ledger/audit log and notify sender and recipient.

UML DIAGRAM

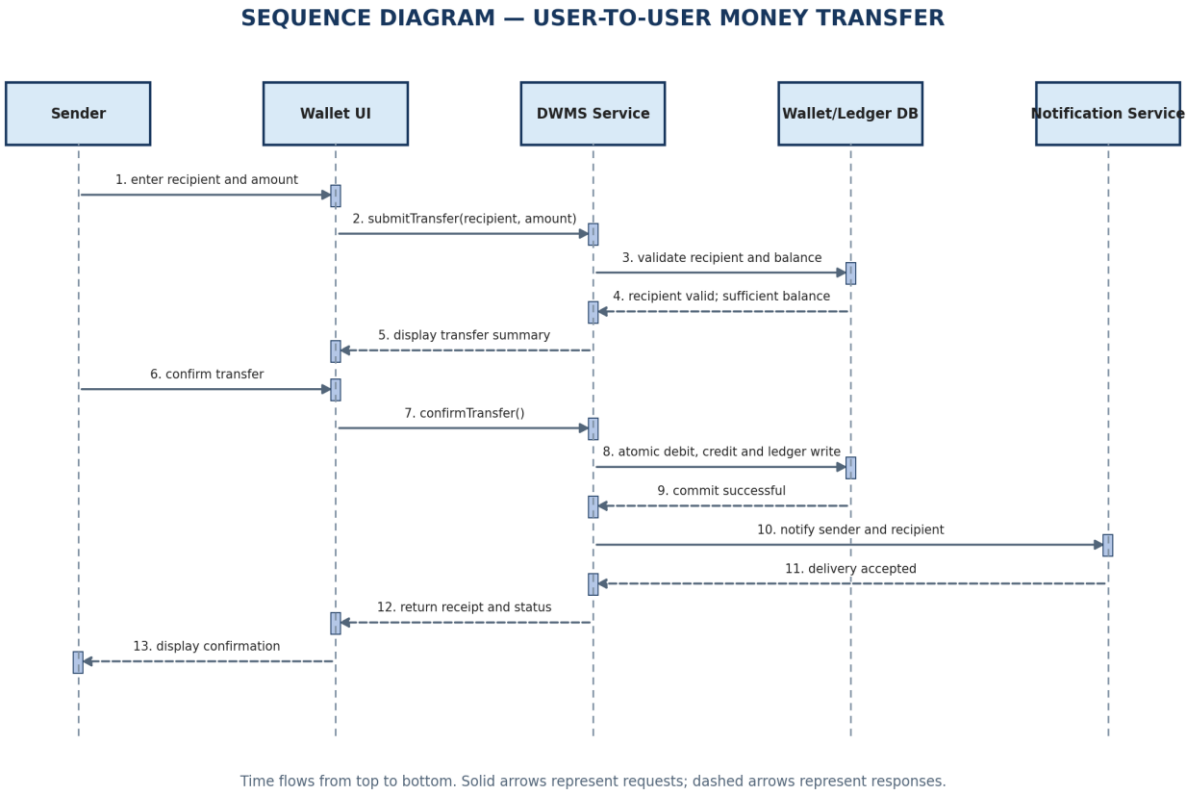


Figure 1. Sequence diagram for the money-transfer scenario.

3. Message Flow

1. The sender enters the recipient and amount in the Wallet UI.
2. The UI submits the transfer request to the DWMS service.
3. The service checks recipient validity and sender balance using the persistent data layer.
4. After validation, the UI displays the transfer summary and requests explicit confirmation.
5. The service commits the debit, credit, and ledger record as one atomic operation.
6. After a successful commit, the service requests notifications for both parties and returns a receipt/status to the UI.

Conclusion

The sequence diagram shows that validation and confirmation precede the financial commit. The atomic database operation is central to avoiding inconsistent wallet balances, while the receipt and notifications provide user feedback and traceability. The model can be used to derive tests for invalid recipients, insufficient funds, cancelled transfers, commit failure, and successful transfer.